

# DW\_decode\_en

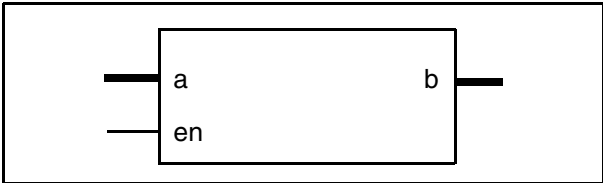
## Binary Decoder with Enable

Version, STAR and Download Information: [IP Directory](#)

### Features and Benefits

- Parameterized word length
- Integrates enable control

### Revision History



### Description

DW\_decode\_en decodes an address on input port *a* to a single bit-line on output port *b*. A decoder with *width* = *n* bits has  $2^n$  bits at the output, where each output with index *i* becomes active when *en* = 1 (enable input) and input *a* = *i*. The selected output bit is active high. When *en* = 0 none of the output bits are active.

Table 1-1 Pin Description

| Pin Name | Width        | Direction | Function                   |
|----------|--------------|-----------|----------------------------|
| en       | 1 bit        | Input     | Enable input (active high) |
| a        | <i>width</i> | Input     | Binary input data          |
| b        | $2^{width}$  | Output    | Decoded output data        |

Table 1-2 Parameter Description

| Parameter                 | Values  | Description                                                                                      |
|---------------------------|---------|--------------------------------------------------------------------------------------------------|
| <i>width</i> <sup>a</sup> | 1 to 16 | Word length of input <i>a</i> is <i>width</i> .<br>Word length of output <i>b</i> is $2^{width}$ |

a. The *width* parameter value causes the size of output *b* to grow exponentially. Therefore, a *width* value near the upper bound results in a long compile time.

Table 1-3 Synthesis Implementations

| Implementation Name | Function                                | License Feature Required |
|---------------------|-----------------------------------------|--------------------------|
| str                 | Synthesis model                         | DesignWare               |
| cgen                | Datapath generator-based Implementation | DesignWare               |

**Table 1-4 Simulation Models**

| Model                            | Function                             |
|----------------------------------|--------------------------------------|
| DW01.DW_DECODE_EN_CFG_SIM        | Design unit name for VHDL simulation |
| dw/dw01/src/DW_decode_en_sim.vhd | VHDL simulation model source code    |
| dw/sim_ver/DW_decode_en.v        | Verilog simulation model source code |

**Table 1-5 Truth Table (width = 3)**

| A(2:0) | en | B(7) | B(6) | B(5) | B(4) | B(3) | B(2) | B(1) | B(0) |
|--------|----|------|------|------|------|------|------|------|------|
| 000    | 1  | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 1    |
| 001    | 1  | 0    | 0    | 0    | 0    | 0    | 0    | 1    | 0    |
| 010    | 1  | 0    | 0    | 0    | 0    | 0    | 1    | 0    | 0    |
| 011    | 1  | 0    | 0    | 0    | 0    | 1    | 0    | 0    | 0    |
| 100    | 1  | 0    | 0    | 0    | 1    | 0    | 0    | 0    | 0    |
| 101    | 1  | 0    | 0    | 1    | 0    | 0    | 0    | 0    | 0    |
| 110    | 1  | 0    | 1    | 0    | 0    | 0    | 0    | 0    | 0    |
| 111    | 1  | 1    | 0    | 0    | 0    | 0    | 0    | 0    | 0    |
| xxx    | 0  | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    |

## Related Topics

- [Logic – Combinational Overview](#)
- [DesignWare Building Block IP User Guide](#)

## HDL Usage Through Component Instantiation - VHDL

```
library IEEE,DWARE;
use IEEE.std_logic_1164.all;
use DWARE.DWpackages.all;
use DWARE.DW_Foundation_comp_arith.all;

entity DW_decode_en_inst is
    generic (
        inst_width : NATURAL := 8
    );
    port (
        inst_en : in std_logic;
        inst_a : in std_logic_vector(inst_width-1 downto 0);
        b_inst : out std_logic_vector(2**inst_width-1 downto 0)
    );
end DW_decode_en_inst;

architecture inst of DW_decode_en_inst is

begin

    -- Instance of DW_decode_en
    U1 : DW_decode_en
    generic map (
        width => inst_width
    )
    port map (
        en => inst_en,
        a => inst_a,
        b => b_inst
    );

end inst;

-- pragma translate_off
configuration DW_decode_en_inst_cfg_inst of DW_decode_en_inst is
    for inst
    end for; -- inst
end DW_decode_en_inst_cfg_inst;
-- pragma translate_on
```

## HDL Usage Through Component Instantiation - Verilog

```
module DW_decode_en_inst( inst_en, inst_a, b_inst );

parameter inst_width = 8;

input inst_en;
input [inst_width-1 : 0] inst_a;
output [(1<<inst_width)-1 : 0] b_inst;

    // Instance of DW_decode_en
    DW_decode_en #(inst_width) U1 (
        .en(inst_en),
        .a(inst_a),
        .b(b_inst) );

endmodule
```

## Revision History

For notes about this release, see the [DesignWare Building Block IP Release Notes](#).

For lists of both known and fixed issues for this component, refer to the [STAR report](#).

For a version of this datasheet with visible change bars, click [here](#).

| Date           | Release       | Updates                                                                                                                                                   |
|----------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| September 2024 | DWBB_202409.0 | ■ Updated date and version; no technical changes                                                                                                          |
| December 2023  | DWBB_202312.0 | ■ Updated date and version; no technical changes                                                                                                          |
| October 2019   | DWBB_201903.5 | ■ Adjusted the footnote in <a href="#">Table 1-2</a> on page <a href="#">1</a><br>■ Added this Revision History table and the document links on this page |

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